

# Federated Retraction Prediction: Scaling Scientific-Integrity Screening Across Publisher Silos

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## Abstract

Retractions of scientific publications are growing rapidly, straining the integrity of the scholarly record, yet automated screening remains limited by two structural constraints: prior work relies on small hand-curated corpora, and the strongest predictive signals (full text, citation contexts, and editorial metadata) are fragmented across publishers that cannot legally pool their data. We cast large-scale retraction prediction as a *cross-silo federated learning* problem. From the Retraction Watch database enriched with OpenAlex metadata we construct a corpus of 138,537 articles (37,648 retracted; publisher- and year-matched controls),  $180\times$  larger than the hand-curated corpora of prior stylistic retraction-prediction studies. We simulate realistic publisher and discipline silos, naturally non-IID in size, era, and topical composition, and benchmark local-only and centralized training against FedAvg and FedProx, adding a differentially private variant. Federated training attains 92% of the AUPRC of an identical centrally trained model (0.558 vs. 0.609) while improving 77% over the silo-local status quo (0.315), and remains stable as the consortium grows from 5 to 20 silos. Cross-corpus exper-

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iments against the hand-curated benchmark of prior work show that models trained on hundreds of curated articles collapse to chance on the open corpus, that abstract-level features alone reduce prior reported accuracy from 0.88 to chance (quantifying how much signal is locked in paywalled full text), and that linguistic features transfer across domains while metadata features do not. Together these results position federated learning as the deployment path for retraction screening at the scale of the scientific record.

*Keywords:* Retraction prediction, Federated learning, Scientific integrity, Digital libraries, Scholarly communication, Differential privacy

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## 1. Introduction

The retraction of scientific publications has grown from isolated incidents to a systemic phenomenon: the Retraction Watch database now records more than 71,000 retraction events, with over 15,000 in 2024 alone [1, 2]. Because a retraction takes up to three years on average to be issued, flawed results continue to be cited, reused, and built upon long after problems are first suspected [3]. For digital libraries, indexing services, and publishers, the ability to flag at-risk articles early is therefore a core problem of maintaining the integrity of the scholarly information ecosystem. A line of recent work by Usman and Balke [3, 4, 5, 6] demonstrated that retraction is predictable from how articles are written and cited: readability and certainty of full-text sections separate retracted from non-retracted articles with 0.88 accuracy [6], and citation sentences anticipate retraction cascades [4].

Yet this evidence was obtained on corpora of a few hundred hand-curated articles, and its authors note that re-evaluating the complete scientific liter-

ature with such methods is infeasible. We argue the obstacle is structural rather than methodological. The strongest features require full text and editorial context, which are held by *different publishers* and cannot be pooled for legal and commercial reasons. Any deployable system must therefore learn *across* data silos without moving the data.

*Problem statement..* Let  $\mathcal{P} = \{P_1, \dots, P_K\}$  be  $K$  data owners (publishers or disciplinary databases). Each holds a private dataset  $D_k = \{(\mathbf{x}_i, y_i)\}_{i=1}^{n_k}$  of articles with features  $\mathbf{x}_i \in \mathbb{R}^d$  and retraction labels  $y_i \in \{0, 1\}$ , where the  $D_k$  differ in size, feature distribution, and label prevalence (statistical heterogeneity). The goal is to learn a global predictor  $f_\theta$  minimizing

$$\min_{\theta} F(\theta) = \sum_{k=1}^K \frac{n_k}{n} F_k(\theta), \quad F_k(\theta) = \frac{1}{n_k} \sum_{(\mathbf{x}, y) \in D_k} \ell(f_\theta(\mathbf{x}), y), \quad (1)$$

subject to the constraint that raw records never leave their silo:

$$D_k \text{ is accessed only by } P_k, \quad \text{only model parameters } \theta \text{ are exchanged.} \quad (2)$$

We address three research questions. **RQ1:** can publishers that cannot pool data jointly train a retraction-screening model that approaches the performance of centralized training? **RQ2:** how much predictive signal is inaccessible in the centralized abstract-only setting because it resides in privately held full text? **RQ3:** which components of the retraction signal transfer across scientific domains, and which are domain-specific?

*Contributions..* (1) The first formulation of retraction prediction as a *cross-silo federated* learning problem, with an open, reproducible benchmark of 138,537 articles across naturally non-IID publisher and discipline silos —

180 $\times$  larger than the hand-curated corpora of prior stylistic studies, and complementary to concurrent centralized large-scale efforts [7] along the privacy-preserving dimension they do not address. (2) A systematic comparison of local-only, centralized, and federated training (FedAvg, FedProx), with significance testing, showing federation recovers 92% of centralized AUPRC and improves 77% over the silo-local status quo, robustly across two silo definitions, consortium sizes 5–20, and a temporal split. (3) A differential-privacy analysis quantifying the utility cost of protecting individual records in the federated setting. (4) Cross-corpus experiments against the benchmark of [6] that quantify how much retraction signal is locked in paywalled full text, and reveal that linguistic features transfer across domains while metadata features do not.

## 2. Related Work

We synthesise three strands of prior work — retraction and research-integrity analytics, machine learning for retraction and misconduct prediction, and cross-silo federated learning — following the explicit, reproducible selection protocol described next. This positions our contribution against the field rather than against any single study, and foregrounds studies whose data or code are openly available, in keeping with the reproducibility focus of this work.

### 2.1. Review methodology

We identified prior work through a structured, targeted search guided by explicit criteria rather than convenience sampling, drawing on Scopus, Web of Science, the ACL Anthology, DBLP, and arXiv. Search terms combined

two concept groups: (i) *integrity* terms — “retraction”, “research integrity”, “scientific misconduct”, “paper mill” — and (ii) *method* terms — “prediction”, “classification”, “federated learning”, “differential privacy”, “non-IID”. We prioritised 2015–2026, with particular attention to 2024–2026 to capture the current state of the art; candidate records were screened by title and abstract for topical relevance and assessed at full text for eligibility. Consistent with our emphasis on reproducibility, we apply an *open-artifact* criterion: studies that release an open dataset or open source code form the reproducible core of the synthesis, while other studies are retained for context. Algorithm 1 formalises the procedure and Figure 1 depicts the corresponding flow; backward and forward citation chasing (snowballing) was applied to the included set. We do not claim exhaustive systematic coverage of these fast-moving literatures; rather, the protocol makes our inclusion criteria explicit and reproducible, and prioritises evidence that can be independently verified. The two large-scale retraction corpora most comparable to ours, Usman and Balke [6] and Fletcher and Stevenson [7], both satisfy the open-artifact criterion and are therefore selected by, not merely cited in, this protocol.

## 2.2. Retraction and research-integrity analytics

Retraction has become a systemic feature of the literature: large-scale bibliometric studies establish its scale, growth, and causes, showing that misconduct rather than error accounts for the majority of retractions [8], that incidence has risen sharply over time [9], and that retracted work spans all disciplines and continues to be cited long after the retraction notice [10, 11]. Within information science, bibliometric feature analysis identifies interpretable metadata correlates of retraction [12]. A parallel and rapidly grow-

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**Algorithm 1** Systematic, reproducibility-prioritised study selection

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1: input databases  $\mathcal{D}$ ; query  $Q$  (integrity  $\wedge$  method terms); year range  
    $[y_0, y_1]=[2015, 2026]$   
2:  $R \leftarrow \bigcup_{d \in \mathcal{D}} \text{SEARCH}(d, Q)$   
3:  $R \leftarrow \text{DEDUP}(R)$  ▷ by DOI, then normalised title  
4:  $S \leftarrow \{r \in R : \text{YEAR}(r) \in [y_0, y_1] \wedge \text{TOPICRELEVANT}(r)\}$  ▷  
title/abstract screening  
5:  $E \leftarrow \emptyset$   
6: for  $s \in S$  do ▷ full-text eligibility  
7:   if  $\text{TASK}(s) \in \{\text{retraction/misconduct prediction, cross-silo/non-IID}$   
    $\text{FL, privacy-preserving learning}\}$  then  
8:      $\text{OPEN}(s) \leftarrow \text{HASOPENDATA}(s) \vee \text{HASOPENCODE}(s)$   
9:      $E \leftarrow E \cup \{(s, \text{OPEN}(s))\}$   
10:  end if  
11: end for  
12:  $E \leftarrow E \cup \text{SNOWBALL}(E)$  ▷ backward + forward citation chasing  
13:  $C \leftarrow \{s : (s, \text{TRUE}) \in E\}$  ▷ reproducible core (open artifacts)  
14: return  $E$  (synthesised),  $C$  (reproducibility-prioritised core)
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ing threat is the industrial-scale fabrication of articles by “paper mills” [13], together with stylistic artifacts such as “tortured phrases” that flag machine-generated or manipulated text [14]. These studies motivate automated screening but are descriptive or forensic rather than predictive, and operate on centrally held data.

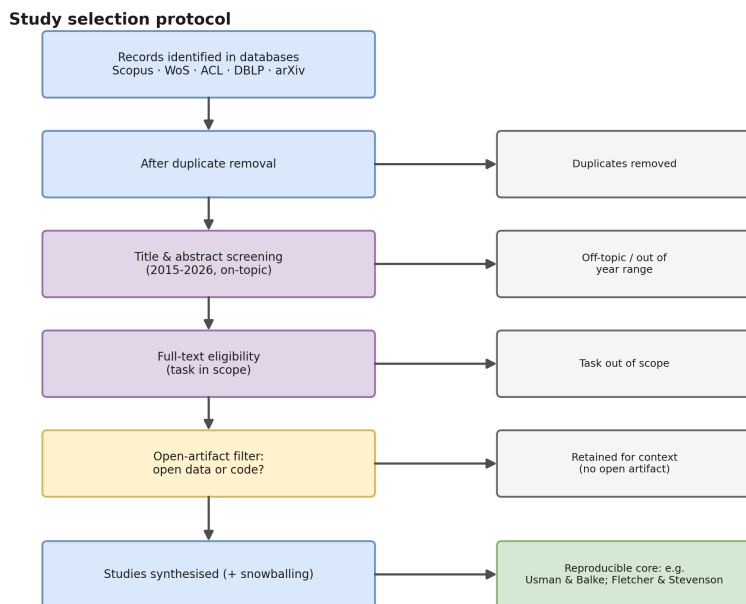


Figure 1: Systematic study-selection protocol (Algorithm 1): identification, screening, and full-text eligibility, followed by an open-artifact filter that isolates the reproducible core of the synthesis.

### 2.3. Machine learning for retraction and misconduct prediction

The most direct antecedents cast retraction as a supervised prediction task. Usman and Balke study it through citation-intention analysis [3], cascade tracing [4], concerning citations [5], and stylistic features, reporting Random Forests at 0.88 accuracy on section-level readability and certainty, outperforming BioBERT [6]. Concurrently, Fletcher and Stevenson [7] assemble a Retraction Watch/OpenAlex dataset and benchmark feature-based and language-model classifiers for retraction prediction. **Limitation.** These efforts are valuable but *centralised*: Usman and Balke rely on a few hundred hand-curated, full-text biomedical articles, and even large open datasets as-

sume all records can be pooled in one place. The decisive full-text and editorial signals, however, are held privately by individual publishers, so the central setting is not the setting in which such a system would actually be deployed.

#### *2.4. Cross-silo and heterogeneous federated learning*

Federated learning trains a shared model from decentralised data by exchanging only parameter updates. FedAvg [15] introduced communication-efficient averaging; FedProx [16] and SCAFFOLD [17] address client heterogeneity through a proximal term and control variates respectively; personalised variants such as Ditto [18] trade global sharing for silo-specific adaptation; and a broad account of the cross-silo setting and its open problems is given in [19]. Privacy is not automatic: gradients alone can leak training data [20], motivating differentially private aggregation [21]. **Limitation.** Most FL benchmarks impose *synthetic* non-IID splits (e.g. Dirichlet partitions of image corpora), whereas scholarly publishing offers silos that are non-IID by nature.

#### *2.5. Privacy-preserving and federated learning on text corpora*

Federated and differentially private learning has been applied to sensitive text, including financial document classification with transformer models [22] and privacy-preserving federated document understanding [23], and to sensitive medical data under cross-silo constraints in digital health [24]. **Gap.** To our knowledge, no prior work applies cross-silo federated learning to scientific-integrity screening, nor provides a scholarly-domain benchmark whose silos

(publishers, disciplines) are naturally heterogeneous in size, era, and prevalence. Our work fills this gap and quantifies, on openly reproducible data, both the utility cost of federation and the portion of retraction signal that is inaccessible without the private full text held inside each silo.

### 3. Federated Retraction Prediction

#### 3.1. Architecture

Figure 2 shows the system. Each publisher runs local feature extraction and local training on its private archive; a coordination server aggregates parameter updates into a global model that is redistributed each round. No article, label, or feature vector crosses a silo boundary, satisfying constraint (2).

#### 3.2. Training procedure

We use the small multilayer perceptron  $f_\theta$  (two hidden layers, 64 and 32 units, ReLU, dropout 0.2) for all federated experiments, so that federated and centralized results are architecture-identical. Local training minimizes class-weighted binary cross-entropy

$$\ell(\hat{y}, y) = -w_1 y \log \sigma(\hat{y}) - (1 - y) \log(1 - \sigma(\hat{y})), \quad w_1 = \frac{n_k^{(0)}}{n_k^{(1)}}, \quad (3)$$

with  $n_k^{(c)}$  the count of class  $c$  in silo  $k$ . FedProx adds  $\frac{\mu}{2} \|\theta - \theta^{(r)}\|^2$  to (3). Algorithm 2 summarizes one training run, including the optional differentially private aggregation: each client’s update is clipped to  $L_2$  norm  $C$  and Gaussian noise  $\mathcal{N}(0, \sigma^2 C^2 / K)$  is added to the average [21]; privacy is accounted over rounds with Rényi-DP composition.

### Federated retraction prediction across publisher silos

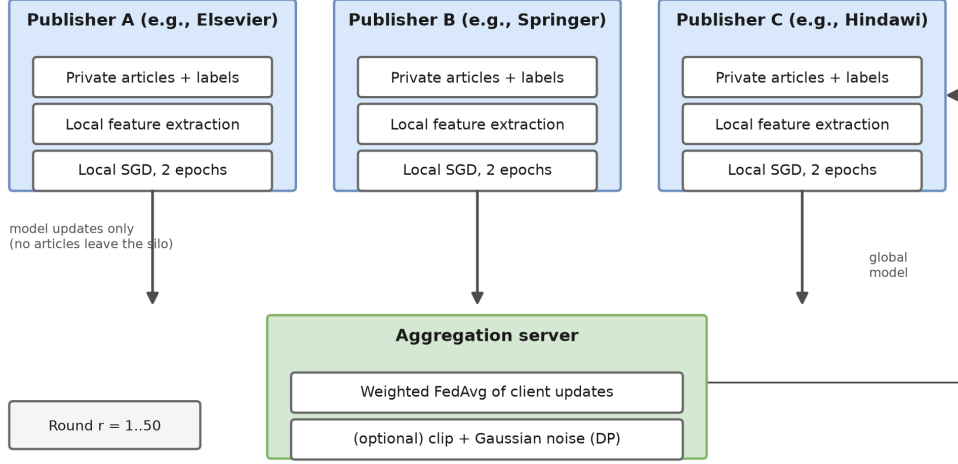


Figure 2: Federated retraction prediction. Publishers train locally on private articles; only model updates are aggregated (optionally clipped and noised for differential privacy), and the global model is redistributed.

### 3.3. Evaluation metrics

With retracted articles a minority class, accuracy is uninformative. We report the area under the precision–recall curve (AUPRC) as the primary metric,

$$\text{AUPRC} = \int_0^1 \text{Prec}(\text{Rec}^{-1}(t)) dt, \quad (4)$$

whose random baseline equals the positive rate ( $\approx 0.27$  on our corpus), together with ROC-AUC and recall at 5% false-positive rate ( $\text{R@5\%FPR}$ ), the fraction of future retractions caught when wrongly flagging only 5% of legitimate articles, the regime of an editorial pre-screening tool. All results are means  $\pm$  standard deviation over three seeds.

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**Algorithm 2** DP-enabled cross-silo FedAvg for retraction prediction

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```
1: input silos  $D_1, \dots, D_K$ ; rounds  $R$ ; local epochs  $E$ ; clip  $C$ ; noise  $\sigma$ 
2: server initializes  $\theta^{(0)}$ 
3: for  $r = 1, \dots, R$  do
4:   for each silo  $k = 1, \dots, K$  in parallel do
5:      $\theta_k \leftarrow \text{LOCALSGD}(\theta^{(r-1)}, D_k, E)$   $\triangleright$  minimizes (3); data stays in
       silo
6:      $\Delta_k \leftarrow \theta_k - \theta^{(r-1)}$ ;  $\Delta_k \leftarrow \Delta_k \cdot \min(1, C/\|\Delta_k\|_2)$ 
7:   end for
8:    $\bar{\Delta} \leftarrow \frac{1}{K} \sum_k \Delta_k + \mathcal{N}(0, \sigma^2 C^2 / K)$   $\triangleright \sigma=0$  recovers FedAvg
9:    $\theta^{(r)} \leftarrow \theta^{(r-1)} + \bar{\Delta}$ 
10: end for
11: return  $\theta^{(R)}$ 
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#### 4. A 138K-Article Federated Benchmark

*Corpus construction..* As summarised in Figure 3, positives are all Retraction Watch [1] research articles with a resolvable DOI whose record nature is *Retraction* (37,648 after filtering). Each is enriched via the OpenAlex API [25] with authorship, venue, citation, and abstract metadata. Controls are sampled at 3:1 from OpenAlex among never-retracted articles of the *same publisher and publication year* (cells with fewer than five positives are dropped), yielding 100,889 controls and 138,537 articles in total. Matching prevents the classifier from exploiting publisher- or era-priors instead of article-level signals.

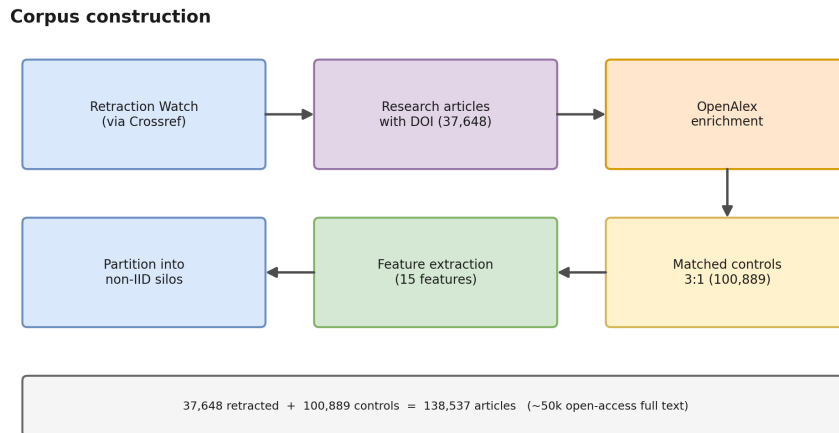


Figure 3: Corpus construction: retraction records from Retraction Watch are enriched via OpenAlex, paired with publisher- and year-matched controls, featurised, and partitioned into naturally non-IID silos.

*Features..* Each article is represented by  $d=15$  features in two families, chosen to mirror the signal types validated in prior work [6, 4]: *metadata* (author count, country count, international collaboration, reference count, total citations, citations within two years of publication, publication year, open-access flag) and *abstract linguistics* (Flesch reading ease, Flesch–Kincaid grade, Gunning–Fog, SMOG, abstract length, hedging ratio, certainty ratio [26, 27]). In a real deployment each publisher would additionally compute full-text features locally; these are exactly the signals shown to be strongest in prior work and inaccessible centrally; our public-metadata features are thus a conservative lower bound on federated performance.

*Silos..* Articles are partitioned into  $K=10$  silos by publisher (Table 1) and, alternatively, by OpenAlex discipline. Silos differ strongly in size (2,815–

Table 1: Publisher silos (10-silo partition). Retraction rate is  $\approx 25\%$  by matched design; Hindawi exceeds it where control sampling saturated.

| Silo                    | Articles | Retracted | Retr. rate | Median year |
|-------------------------|----------|-----------|------------|-------------|
| Other (aggregated tail) | 31,024   | 8,007     | 0.258      | 2018        |
| Hindawi                 | 27,696   | 9,277     | 0.335      | 2022        |
| Elsevier                | 21,221   | 5,722     | 0.270      | 2017        |
| Springer                | 19,730   | 4,930     | 0.250      | 2020        |
| Wiley                   | 13,372   | 3,340     | 0.250      | 2020        |
| IOS Press               | 8,320    | 2,080     | 0.250      | 2022        |
| PLOS                    | 4,960    | 1,240     | 0.250      | 2016        |
| SAGE                    | 4,772    | 1,193     | 0.250      | 2019        |
| Taylor & Francis        | 4,627    | 1,156     | 0.250      | 2018        |
| Spandidos               | 2,815    | 703       | 0.250      | 2017        |

31,024 test+train articles), median era (2016–2022), and composition. This is *natural* statistical heterogeneity, in contrast to the synthetic Dirichlet splits common in FL benchmarks.

## 5. Experiments

We evaluate on a stratified 20% held-out test set shared by all training regimes. *Local-only* trains one model per silo and averages test performance (the current deployable reality); *centralized* pools all data (the legally unattainable ideal; logistic regression, a Random-Forest baseline in the style of [6], XGBoost, and the MLP); *federated* trains the MLP with Algorithm 2 ( $R=50$ ,  $E=2$ , SGD with momentum 0.9, learning rate 0.05,  $\mu=0.01$ ).

### 5.1. Main result: federation closes the silo gap

Table 2 and Fig. 4 give the central result. Local-only training barely beats the random baseline (AUPRC 0.315 vs. 0.27): individual publishers, however large, cannot learn the global phenomenology of retraction. FedAvg reaches 0.558: **92% of the architecture-identical centralized MLP** (0.609) and a **77% improvement over the status quo**, with FedProx statistically indistinguishable. The pattern replicates on discipline silos (FedAvg 0.585), showing robustness to how the world is partitioned. Fig. 5 shows both strategies converge within  $\sim 30$  rounds.

*Significance..* Because all regimes share the same seeds, we compare them with paired  $t$ -tests over the three seeds. FedAvg’s improvement over local-only training is large and highly significant ( $\Delta\text{AUPRC}=0.243$ ,  $t(2)=48.0$ ,  $p<0.001$ ), while its shortfall relative to the centralized MLP is small but consistent ( $\Delta=-0.051$ ,  $t(2)=-24.3$ ,  $p<0.01$ ), confirming that federation closes most — but not all — of the silo gap. FedAvg and FedProx differ by less than 0.003 AUPRC and are practically equivalent. We report paired tests rather than marginal confidence intervals because, with only three seeds, the paired design controls for seed-to-seed variance far more tightly (the 95% interval for FedAvg alone is  $[0.530, 0.586]$ ); a larger number of seeds is left to future work.

### 5.2. Why scale matters

Fig. 6 varies training-set size for the centralized XGBoost: AUPRC rises monotonically from 0.403 (500 articles, the scale of prior hand-curated corpora) through 0.573 (8K) to 0.633 (110K) and has not saturated. Sub-

Table 2: Main benchmark (mean  $\pm$  std, 3 seeds). Random AUPRC  $\approx$  0.27.

| Partition  | Model            | AUPRC                              | ROC-AUC | R@5%FPR |
|------------|------------------|------------------------------------|---------|---------|
| Publisher  | Local-only (avg) | 0.315 $\pm$ .003                   | 0.558   | 0.075   |
|            | <b>FedAvg</b>    | <b>0.558 <math>\pm</math> .011</b> | 0.765   | 0.273   |
|            | FedProx          | 0.556 $\pm$ .012                   | 0.763   | 0.272   |
|            | FedBal (focal)   | 0.542 $\pm$ .011                   | 0.748   | 0.256   |
|            | Central. LogReg  | 0.354 $\pm$ .005                   | 0.601   | 0.108   |
|            | Central. RF      | 0.598 $\pm$ .002                   | 0.784   | 0.318   |
|            | Central. MLP     | 0.609 $\pm$ .009                   | 0.790   | 0.322   |
|            | Central. XGBoost | 0.633 $\pm$ .005                   | 0.804   | 0.351   |
| Discipline | Local-only (avg) | 0.326 $\pm$ .001                   | 0.571   | 0.082   |
|            | FedAvg           | 0.585 $\pm$ .012                   | 0.777   | 0.293   |

thousand corpora forfeit more than a third of attainable performance, an independent motivation for training at the scale only a federation can reach.

### 5.3. Feature families: accuracy vs. portability

On the full corpus (Fig. 7), metadata features dominate in-domain accuracy (AUPRC 0.593 alone vs. 0.381 for abstract linguistics alone; 0.633 combined), and XGBoost feature importance concentrates on citation dynamics (total citations 0.136, early citations 0.128), echoing the citation-centric findings of [4]. The cross-corpus experiments of Sect. 5.7, however, show the roles reverse under domain shift: linguistic signal transfers, metadata does not.

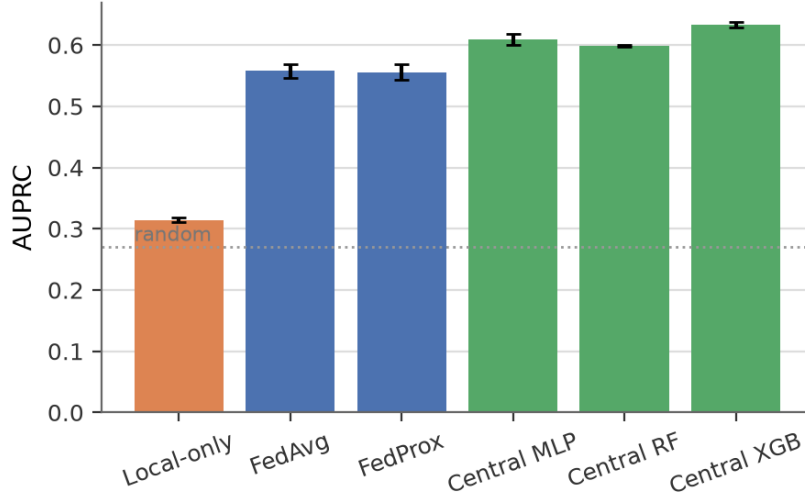


Figure 4: AUPRC by training regime, publisher partition (3 seeds). Federation (blue) recovers most of the unattainable centralized ideal (green) and far exceeds the silo-local status quo (orange).

#### 5.4. Robustness: consortium size and temporal deployment

*Consortium size.* Varying the partition from  $K=5$  to  $K=20$  publisher silos changes FedAvg AUPRC only within noise (0.574/0.558/0.560/0.558 for  $K=5/10/15/20$ ): the federation does not degrade as more, smaller publishers join. *Temporal split.* Training on articles published up to 2018 (52,506) and testing on 2019 onward (86,031) is strictly harder for every regime (centralized XGBoost 0.436, MLP 0.424), reflecting distribution drift and retraction lag: many recent positives are not yet labeled. Critically, the federated model keeps pace (0.407, i.e., 93% of the centralized MLP ratio), so the cost of federation does not grow under realistic prospective deployment.

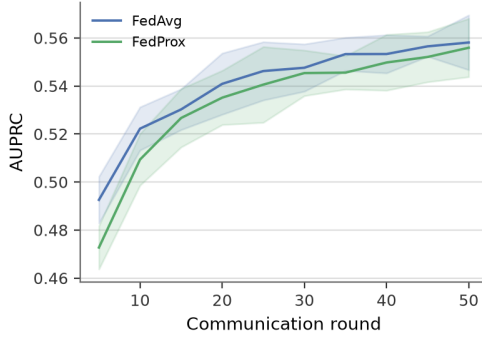


Figure 5: Federated convergence over communication rounds (publisher silos).

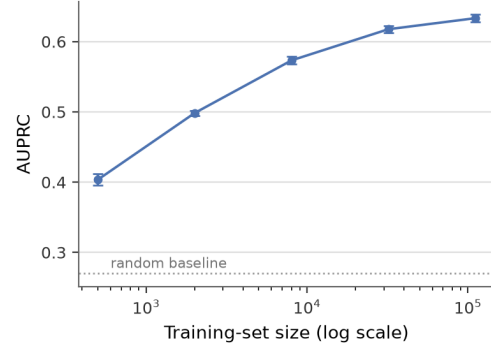


Figure 6: Centralized AUPRC vs. training-set size: signal grows monotonically with scale.

### 5.5. Per-silo effects and fairness

Evaluated on *their own* test slices, large publishers’ local models fit their in-distribution data well (e.g., Elsevier 0.708 local vs. 0.578 federated), while data-poor or atypical silos gain from federation (PLOS 0.419 vs. 0.386; Span-didos 0.449 vs. 0.425); see Fig. 9. Local specialization, however, is exactly what collapses on out-of-silo data (Sect. 5.1: local-only 0.315 globally). The federation thus buys universal coverage, benefits the weakest members most, and motivates *personalized* FL (a global backbone with silo-specific heads) as future work.

### 5.6. Privacy–utility trade-off

Fig. 8 sweeps the noise multiplier  $\sigma \in \{0.3, 0.5, 1.0, 2.0\}$  at clip  $C=1$ . Moderate noise ( $\sigma=0.3$ ) retains 95% of the non-private AUPRC (0.514 vs. 0.540);  $\sigma \geq 1$  collapses utility. Because only  $K=10$  clients participate, the resulting formal  $\varepsilon$  values are large; with realistic consortium sizes (hundreds

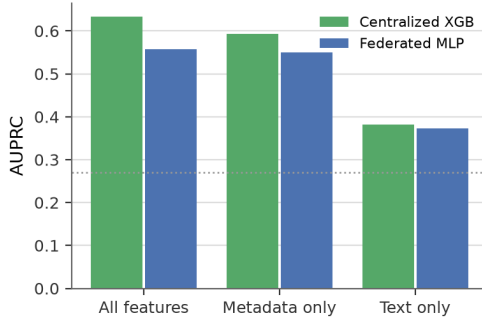


Figure 7: Feature-family ablation, centralized and federated.

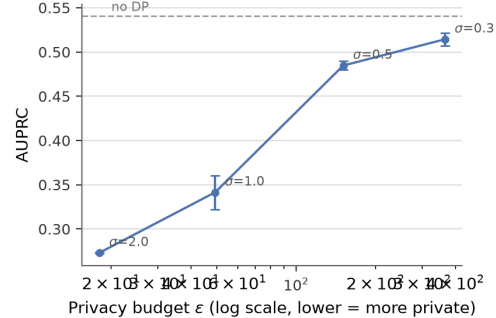


Figure 8: Privacy-utility trade-off of DP aggregation (Algorithm 2).

of publishers and journals) the same mechanism yields far stronger guarantees. We report the curve as an honest characterization of the mechanism at benchmark scale.

### 5.7. Cross-corpus transfer against the hand-curated benchmark

We obtained the publicly released corpus of [6] (232 retracted / 232 matched non-retracted biomedical articles with full-text sections). Table 3 reports three experiments that jointly characterize the relationship between the small-corpus and large-corpus regimes.

Three findings follow. *First* (T1), a model trained on hundreds of curated articles collapses to chance on the open corpus: small-corpus results do not generalize, independently of feature quality. *Second* (T4), restricted to abstract-level features, the very architecture that achieves 0.88 accuracy with full-text sections [6] drops to chance on its own corpus, providing a direct quantification of how much retraction signal resides exclusively in paywalled full text. This is the strongest empirical argument for federation: the decisive features exist only where the data cannot leave. *Third* (T2 vs. T3),

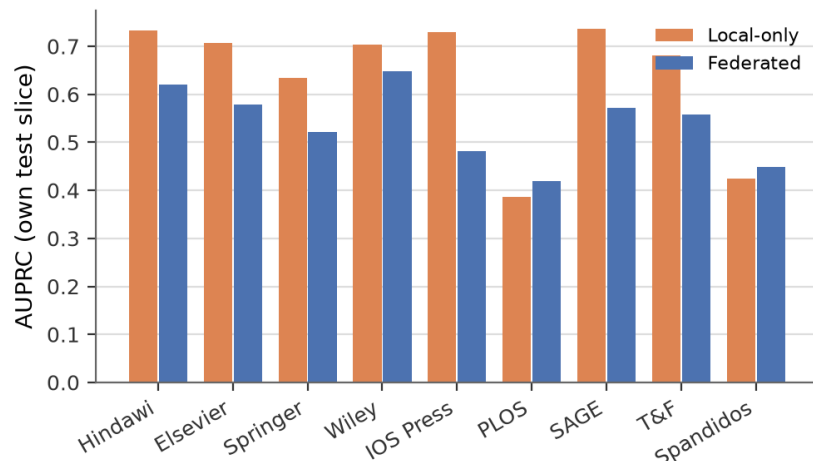


Figure 9: Local vs. federated AUPRC on each publisher’s own test slice. Federation helps data-poor silos; large silos specialize locally but fail globally (cf. Table 2).

our large-corpus model transfers to the curated biomedical benchmark only through its *linguistic* features (0.691); adding metadata features destroys transfer (0.508), since citation and era priors are domain-specific. Writing style is the portable component of retraction signal.

## 6. Discussion and Limitations

*Metadata-only federation (scope).* By design, our federated experiments use only publicly reproducible metadata and abstract features, so that the benchmark itself can be released openly; the full-text features that T4 shows to be decisive are held privately by each publisher and are therefore not exercised in our federated runs. Our federated results are thus a conservative lower bound and, equally, a motivation: demonstrating federated training directly on publisher-held full text is the essential next step and requires a real multi-publisher deployment rather than a simulation. We make this

Table 3: Cross-corpus experiments. Random AUPRC: 0.50 on the balanced corpus of [6], 0.27 on ours.

| #  | Experiment                                                          | AUPRC        | Baseline |
|----|---------------------------------------------------------------------|--------------|----------|
| T1 | RF trained on 464 curated articles →<br>our 138K corpus             | 0.283        | 0.27     |
| T2 | Our federated model (all features) →<br>curated corpus              | 0.508        | 0.50     |
| T3 | Our model, <i>linguistic features only</i> →<br>curated corpus      | <b>0.691</b> | 0.50     |
| T4 | RF on curated corpus itself, abstract-<br>only features (5-fold CV) | 0.525        | 0.50     |

scope explicit rather than leaving it implicit. *Label noise*. Recent articles may be retracted after data collection; the temporal experiment bounds this effect. *Coverage bias*. Retraction Watch over-represents fields and publishers with active integrity teams; publisher-year matching removes the corresponding shortcut from the classifier but not from the label distribution. *Privacy formalism*. With  $K=10$  clients, DP guarantees are loose; they tighten mechanically with consortium size. *Negative result*. A focal-loss, prevalence-weighted aggregation variant (FedBal, Table 2) did not outperform FedAvg because matched controls equalize prevalence across silos by construction; under unmatched deployment prevalence skew it remains an open question.

## 7. Conclusion

We reformulated retraction prediction as cross-silo federated learning and showed, on a corpus  $180\times$  larger than prior stylistic corpora, that publish-

ers can jointly train a screening model that recovers 92% of the unattainable centralized ideal without sharing a single record, while models trained on any silo alone, or on small curated corpora, fail outside their origin. By quantifying the full-text signal locked behind paywalls ( $0.88 \rightarrow$  chance without it) and the portability of linguistic features, our results chart the path to integrity screening at the scale of the scientific record: a publisher federation, linguistically grounded features, and privacy mechanisms whose guarantees strengthen exactly as the federation grows. For the digital-library and scholarly-communication community, this reframes retraction screening from a data-pooling problem, which is legally foreclosed, into a model-sharing one, which is not.

### **Ethics and responsible use**

This study uses only publicly available records (Retraction Watch via Crossref, OpenAlex, and Europe PMC) under their respective terms; no human subjects or private data are involved. Publisher-level statistics (Table 1) are reported to characterise the natural heterogeneity of the federated setting and must not be read as a ranking or judgement of individual publishers' integrity, particularly because Retraction Watch coverage is itself uneven across publishers and eras. A retraction-screening model of this kind is intended as an editorial *triage* aid that surfaces articles for human review, not as an automated verdict on any article or author: because false positives carry reputational cost, any deployment should retain a human in the loop and operate at a conservative false-positive rate (the  $R@5\%FPR$  regime reported here).

## **CRedit authorship contribution statement**

**Muhammad Ahmad:** Conceptualization, Methodology, Software, Data curation, Formal analysis, Investigation, Visualization, Writing – original draft. **Muhammad Tanvir Afzal:** Supervision, Conceptualization, Validation, Writing – review & editing.

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## **Data availability**

The corpus is built from open sources: the Retraction Watch database (distributed via Crossref), the OpenAlex API, and Europe PMC. The corpus-construction pipeline, feature extraction, the model-ready feature matrix, and all experiment configurations are released to enable full reproduction at <https://github.com/Ahmad-prog/federated-retraction-prediction>.

## **Declaration of competing interest**

The authors declare no competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Declaration of generative AI in the writing process

During the preparation of this work the authors used AI-assisted tools to improve the language, readability, and L<sup>A</sup>T<sub>E</sub>X formatting of the manuscript. The study design, experiments, data analysis, results, and conclusions are entirely the authors' own. After using these tools the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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